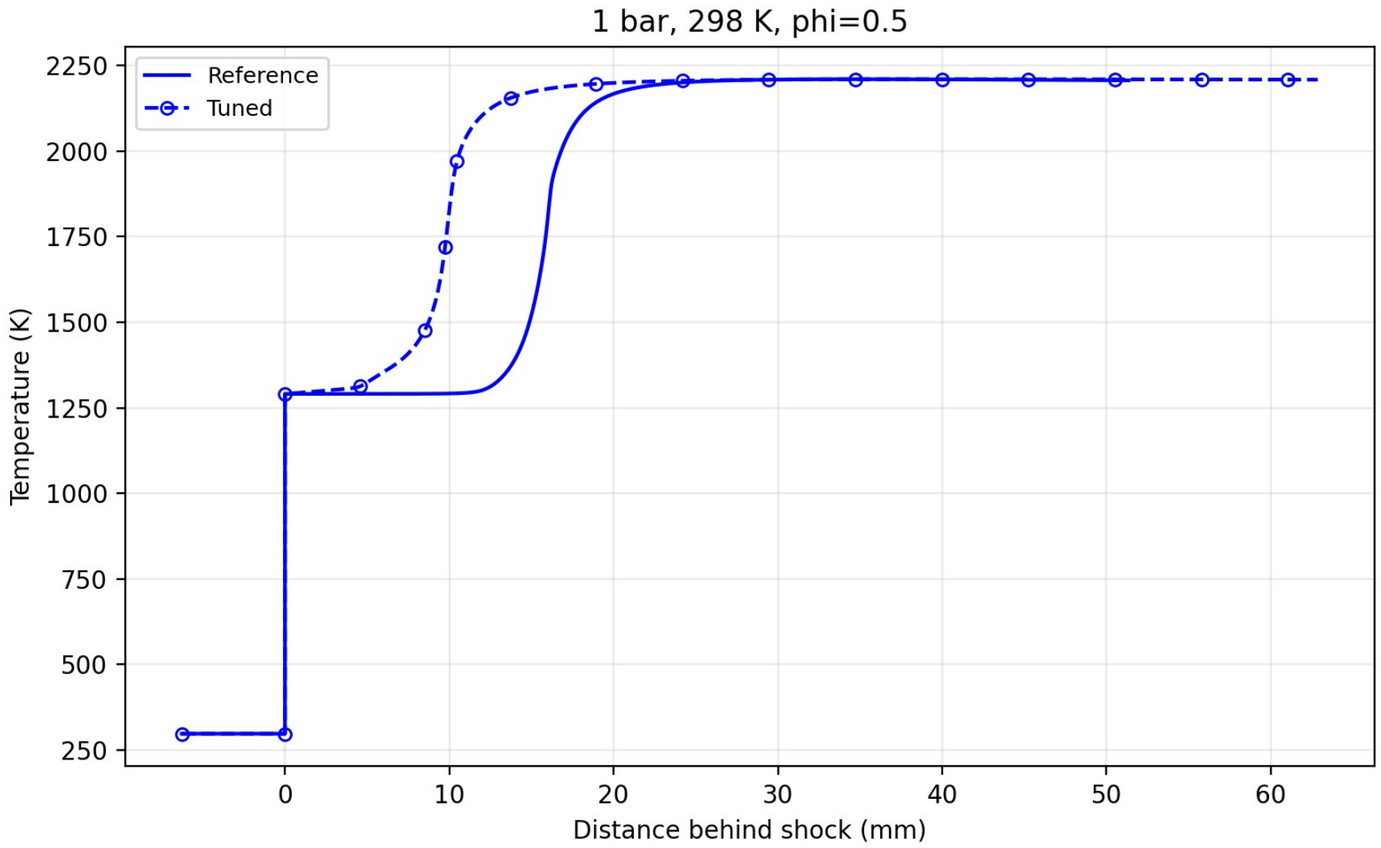
Errors across all 12 conditions

Case	P (bar)	φ	ΔT (K)	Induction error	Exothermic error
1	1	0.5	+1.984	-38.06%	+35.45%
2	1	1	+5.909	-51.43%	-46.34%
3	1	1.5	-8.284	-79.18%	-38.87%
4	1	2	+2.283	-72.68%	-72.97%
5	5	0.5	+10.182	+24.67%	-45.26%
6	5	1	+23.741	-44.29%	-43.84%
7	5	1.5	-12.626	-58.42%	-81.52%
8	5	2	+52.994	-45.40%	-90.97%
9	20	0.5	+1.733	+8.98%	-53.44%
10	20	1	-5.111	-72.93%	-81.51%
11	20	1.5	-5.320	-67.14%	-87.87%
12	20	2	-1.029	-64.26%	-84.09%

Case 1: 1 bar, φ = 0.5

Ethylene–air, initial temperature 298 K



Reported numerical comparison

Quantity	FFCM2	Tuned	Error
T (K)	2206.24	2208.23	+0.09%
ℓind (m)	1.6069e-2	9.9534e-3	-38.06%
ℓexo (m)	7.4218e-4	1.0052e-3	+35.45%

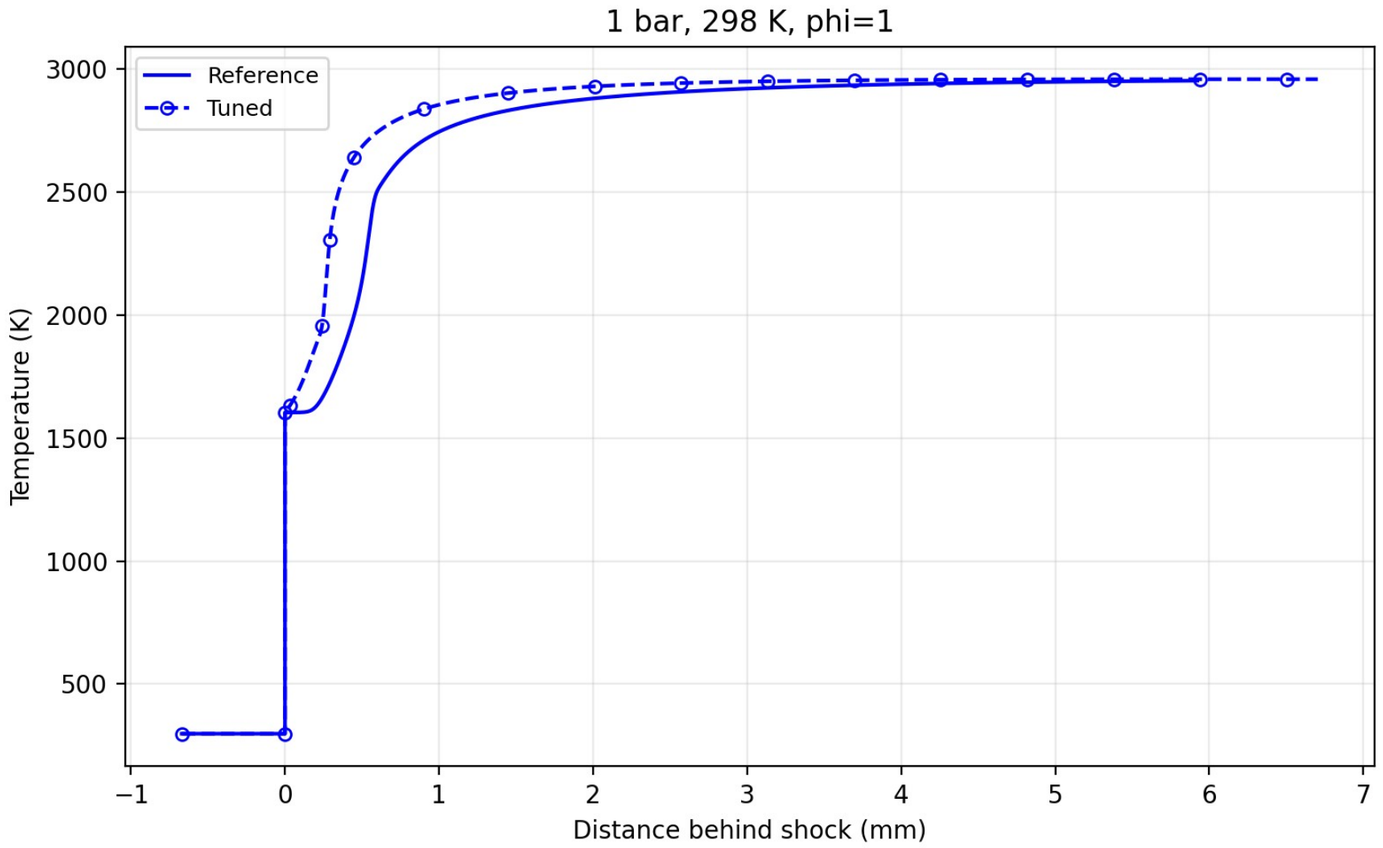
ΔT = +1.984 K

|ln(ℓind,model / ℓind,FFCM2)| = 0.478982

|ln(ℓexo,model / ℓexo,FFCM2)| = 0.303396

Case 2: 1 bar, φ = 1.0

Ethylene–air, initial temperature 298 K



Reported numerical comparison

Quantity	FFCM2	Tuned	Error
T (K)	2952.20	2958.11	+0.20%
ℓind (m)	5.4732e-4	2.6583e-4	-51.43%
ℓexo (m)	1.1799e-4	6.3310e-5	-46.34%

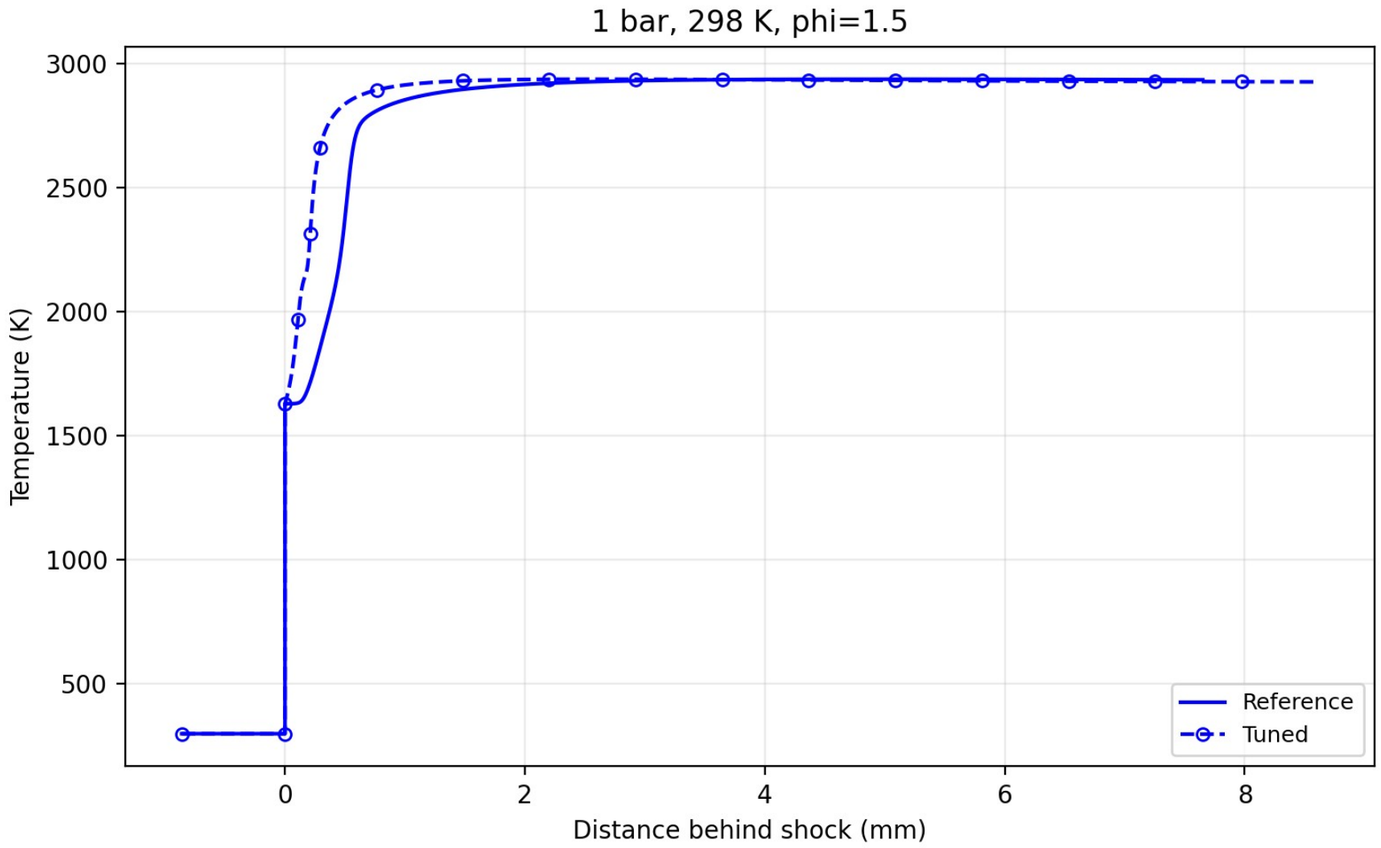
ΔT = +5.909 K

|ln(ℓind,model / ℓind,FFCM2)| = 0.722167

|ln(ℓexo,model / ℓexo,FFCM2)| = 0.622549

Case 3: 1 bar, φ = 1.5

Ethylene–air, initial temperature 298 K



Reported numerical comparison

Quantity	FFCM2	Tuned	Error
T (K)	2933.82	2925.53	-0.28%
ℓind (m)	5.2460e-4	1.0921e-4	-79.18%
ℓexo (m)	1.5716e-4	9.6069e-5	-38.87%

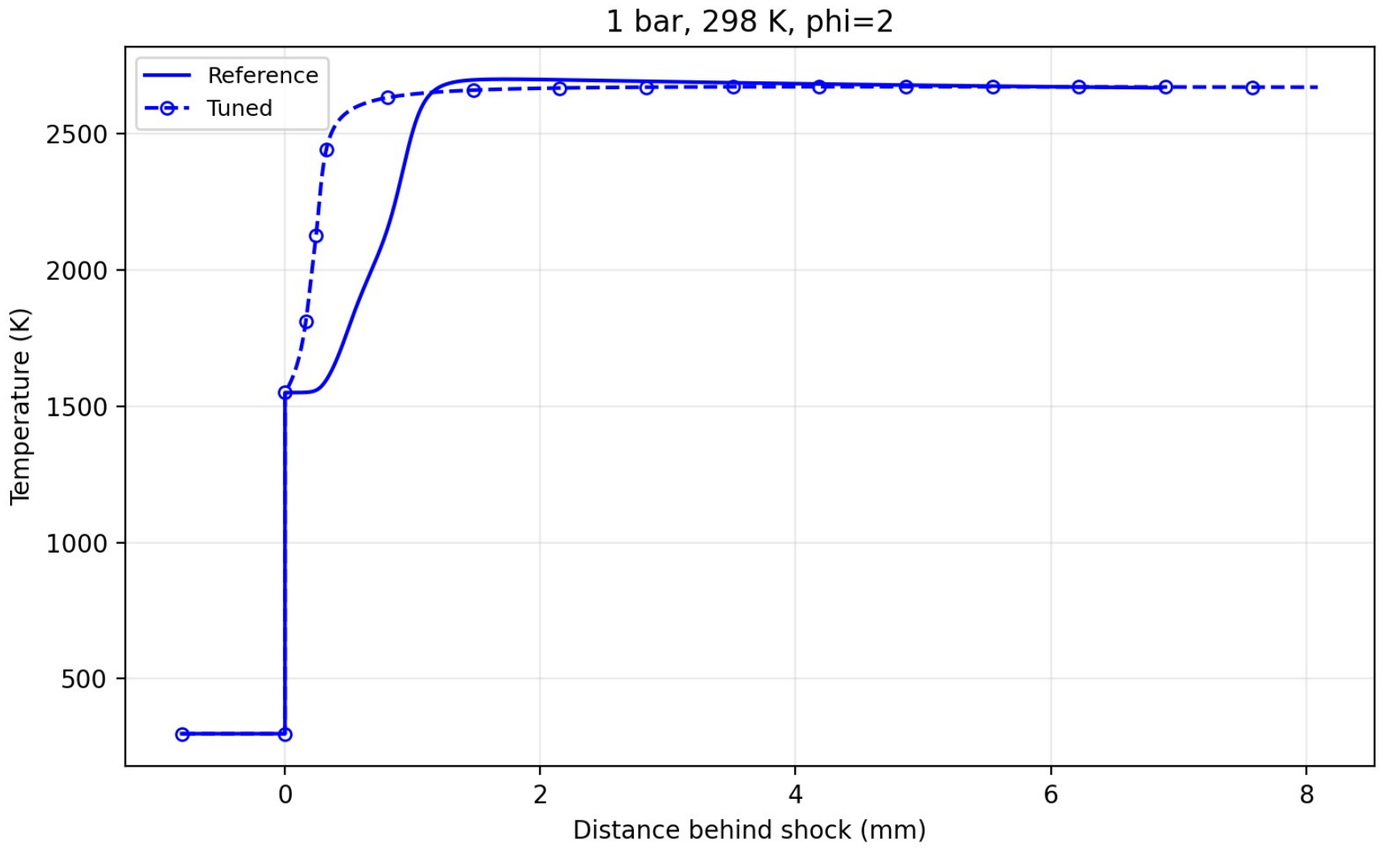
ΔT = -8.284 K

|ln(ℓind,model / ℓind,FFCM2)| = 1.56937

|ln(ℓexo,model / ℓexo,FFCM2)| = 0.492183

Case 4: 1 bar, φ = 2.0

Ethylene–air, initial temperature 298 K



Reported numerical comparison

Quantity	FFCM2	Tuned	Error
T (K)	2668.92	2671.20	+0.09%
ℓind (m)	9.4947e-4	2.5943e-4	-72.68%
ℓexo (m)	6.2757e-4	1.6965e-4	-72.97%

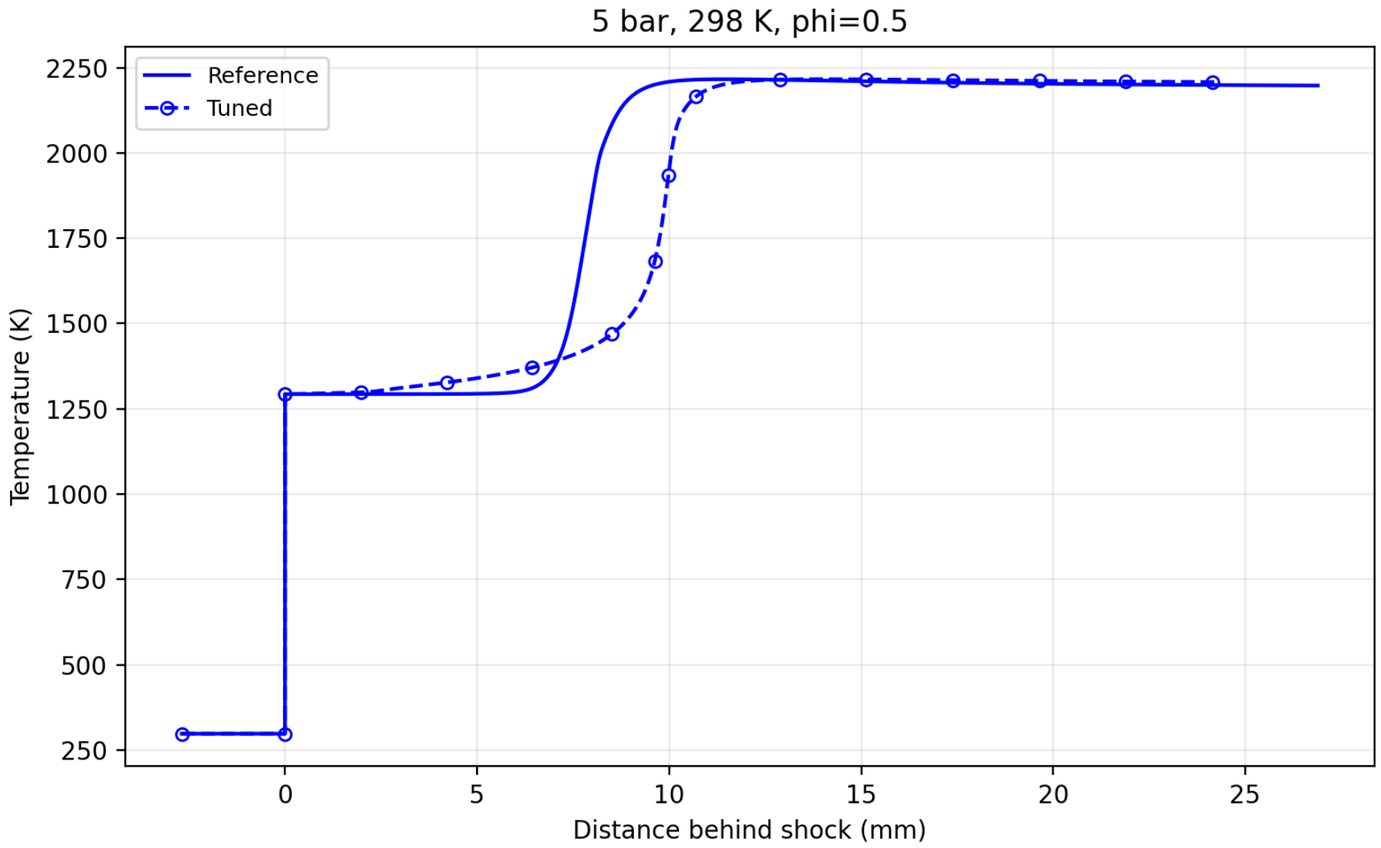
ΔT = +2.283 K

|ln(ℓind,model / ℓind,FFCM2)| = 1.2974

|ln(ℓexo,model / ℓexo,FFCM2)| = 1.30812

Case 5: 5 bar, φ = 0.5

Ethylene–air, initial temperature 298 K



Reported numerical comparison

Quantity	FFCM2	Tuned	Error
T (K)	2197.44	2207.62	+0.46%
ℓind (m)	8.0083e-3	9.9836e-3	+24.67%
ℓexo (m)	8.8781e-4	4.8595e-4	-45.26%

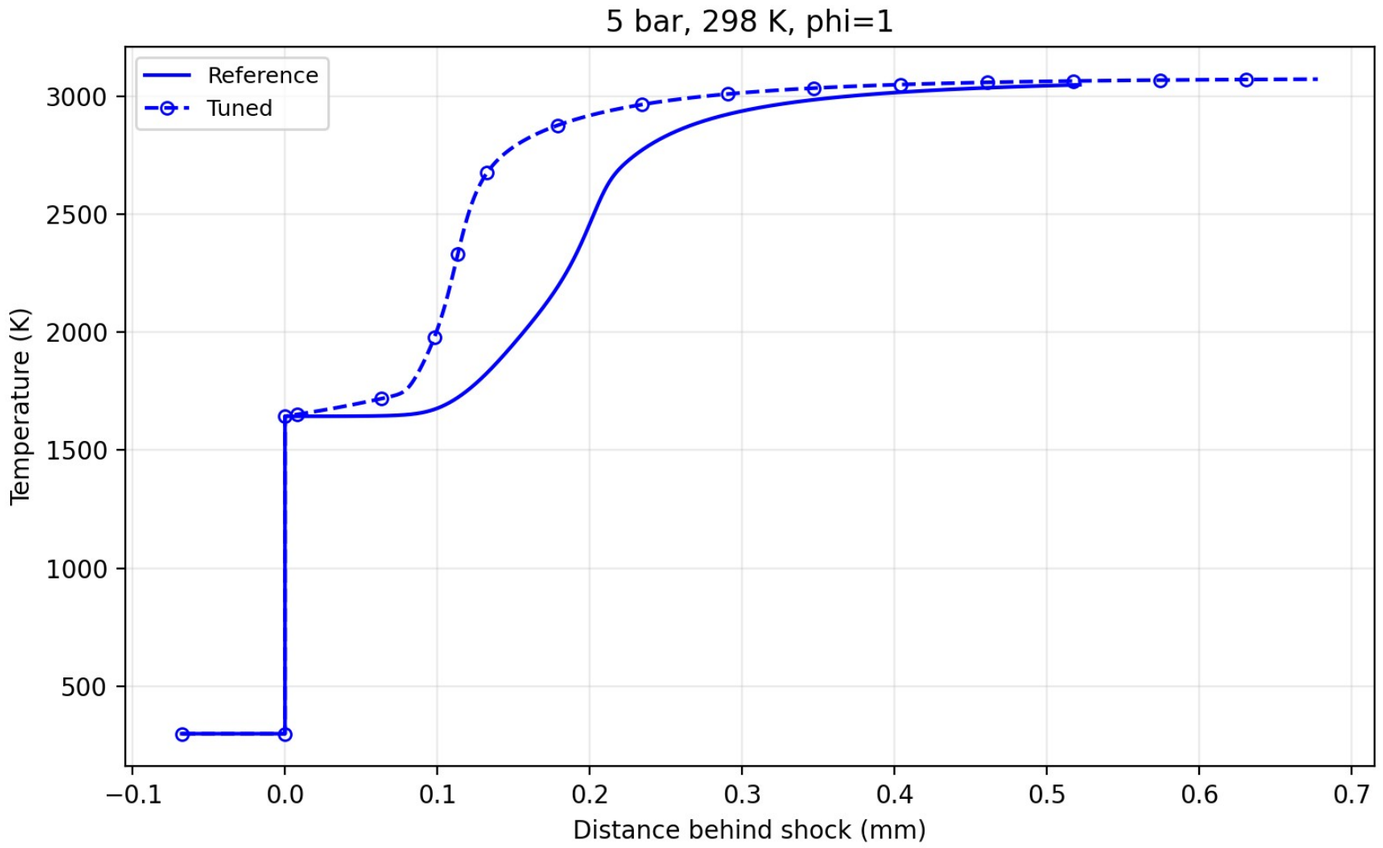
ΔT = +10.182 K

|ln(ℓind,model / ℓind,FFCM2)| = 0.220462

|ln(ℓexo,model / ℓexo,FFCM2)| = 0.602653

Case 6: 5 bar, φ = 1.0

Ethylene–air, initial temperature 298 K



Reported numerical comparison

Quantity	FFCM2	Tuned	Error
T (K)	3049.34	3073.08	+0.78%
ℓind (m)	2.0274e-4	1.1295e-4	-44.29%
ℓexo (m)	5.0900e-5	2.8587e-5	-43.84%

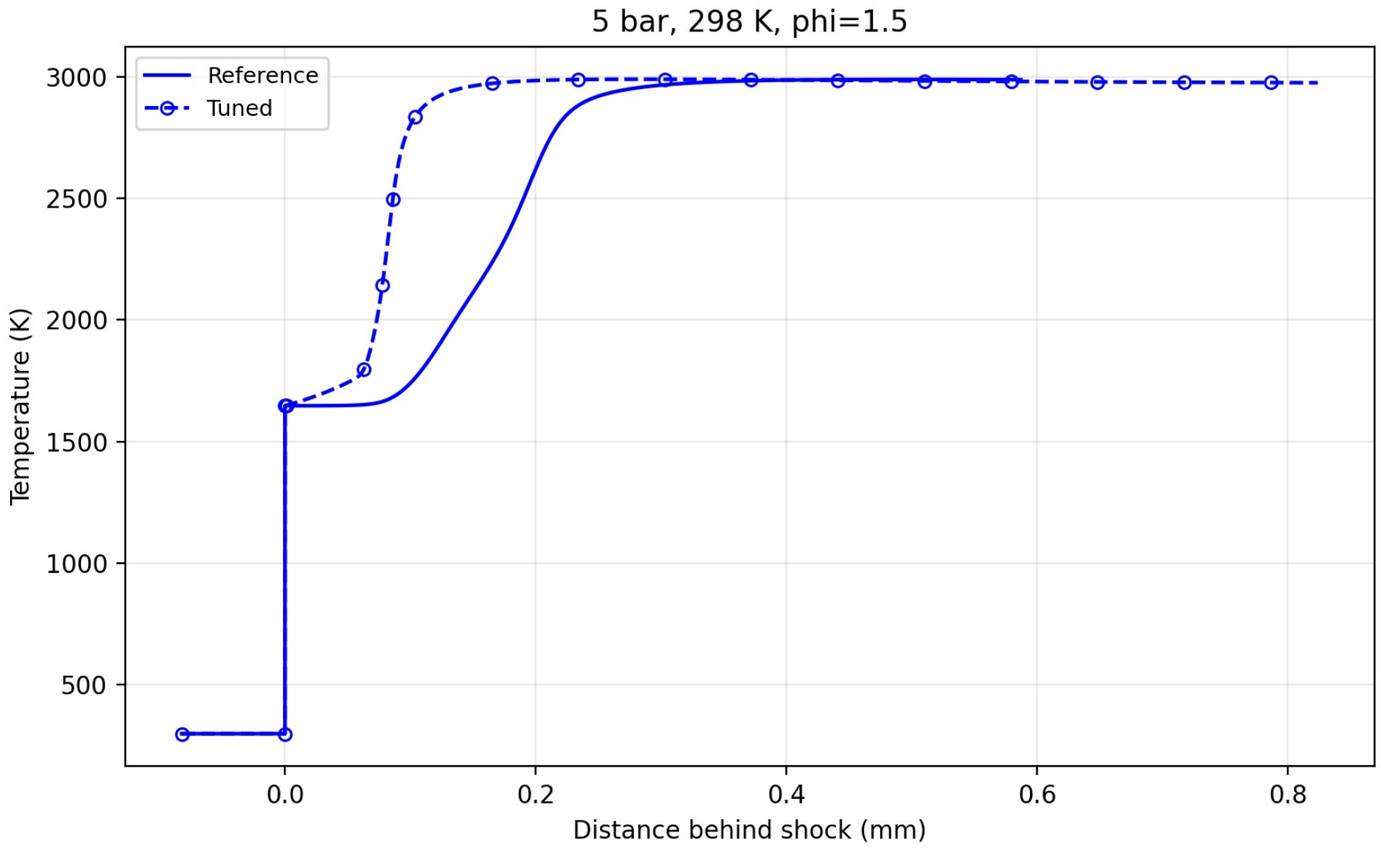
ΔT = +23.741 K

|ln(ℓind,model / ℓind,FFCM2)| = 0.584975

|ln(ℓexo,model / ℓexo,FFCM2)| = 0.576891

Case 7: 5 bar, φ = 1.5

Ethylene–air, initial temperature 298 K



Reported numerical comparison

Quantity	FFCM2	Tuned	Error
T (K)	2987.70	2975.08	-0.42%
ℓind (m)	2.0044e-4	8.3337e-5	-58.42%
ℓexo (m)	1.0433e-4	1.9279e-5	-81.52%

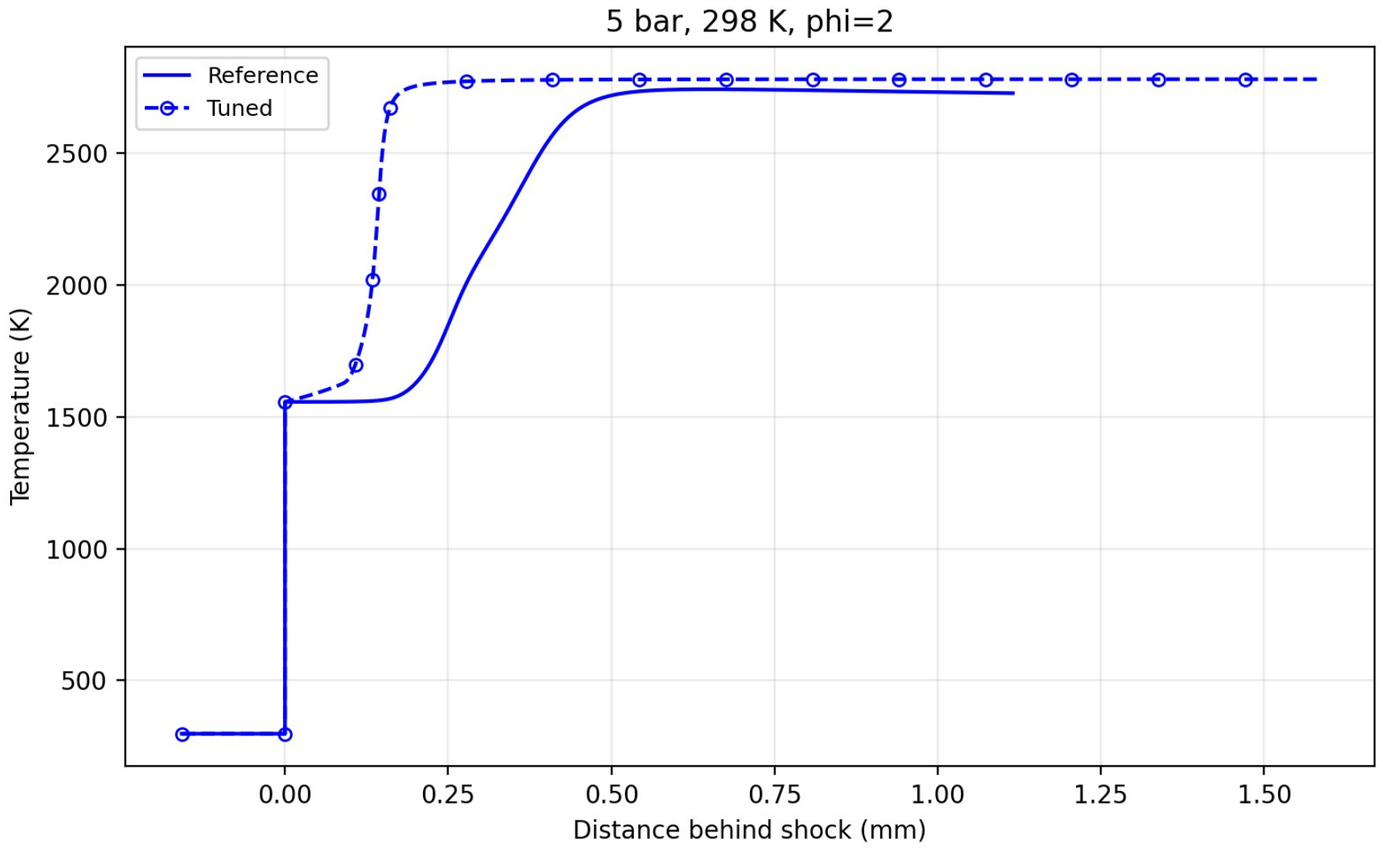
ΔT = -12.626 K

|ln(ℓind,model / ℓind,FFCM2)| = 0.877605

|ln(ℓexo,model / ℓexo,FFCM2)| = 1.68852

Case 8: 5 bar, φ = 2.0

Ethylene–air, initial temperature 298 K



Reported numerical comparison

Quantity	FFCM2	Tuned	Error
T (K)	2727.72	2780.71	+1.94%
ℓind (m)	2.6407e-4	1.4418e-4	-45.40%
ℓexo (m)	2.3135e-4	2.0891e-5	-90.97%

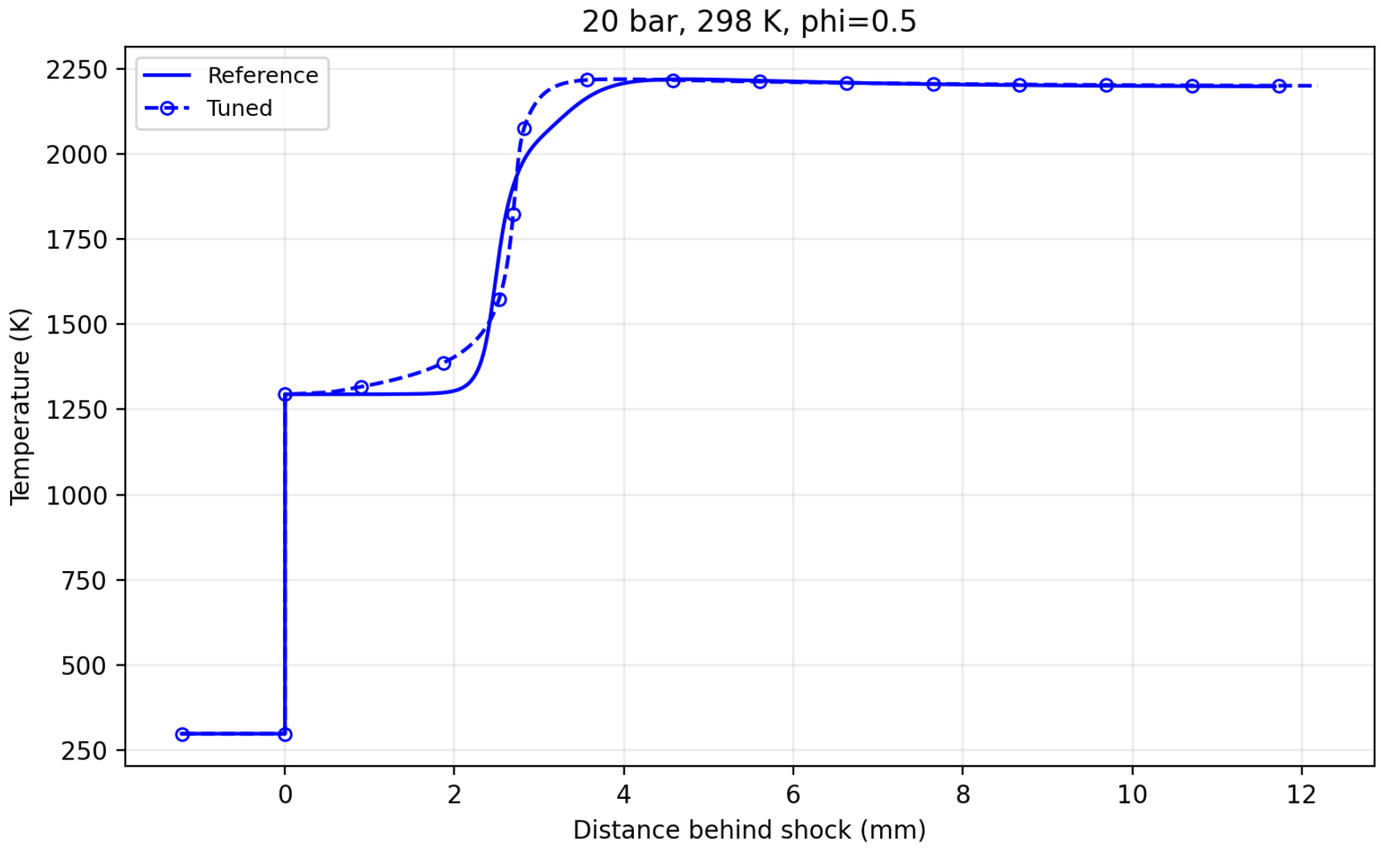
ΔT = +52.994 K

|ln(ℓind,model / ℓind,FFCM2)| = 0.605148

|ln(ℓexo,model / ℓexo,FFCM2)| = 2.4046

Case 9: 20 bar, φ = 0.5

Ethylene–air, initial temperature 298 K



Reported numerical comparison

Quantity	FFCM2	Tuned	Error
T (K)	2198.30	2200.03	+0.08%
ℓind (m)	2.4949e-3	2.7188e-3	+8.98%
ℓexo (m)	3.3413e-4	1.5556e-4	-53.44%

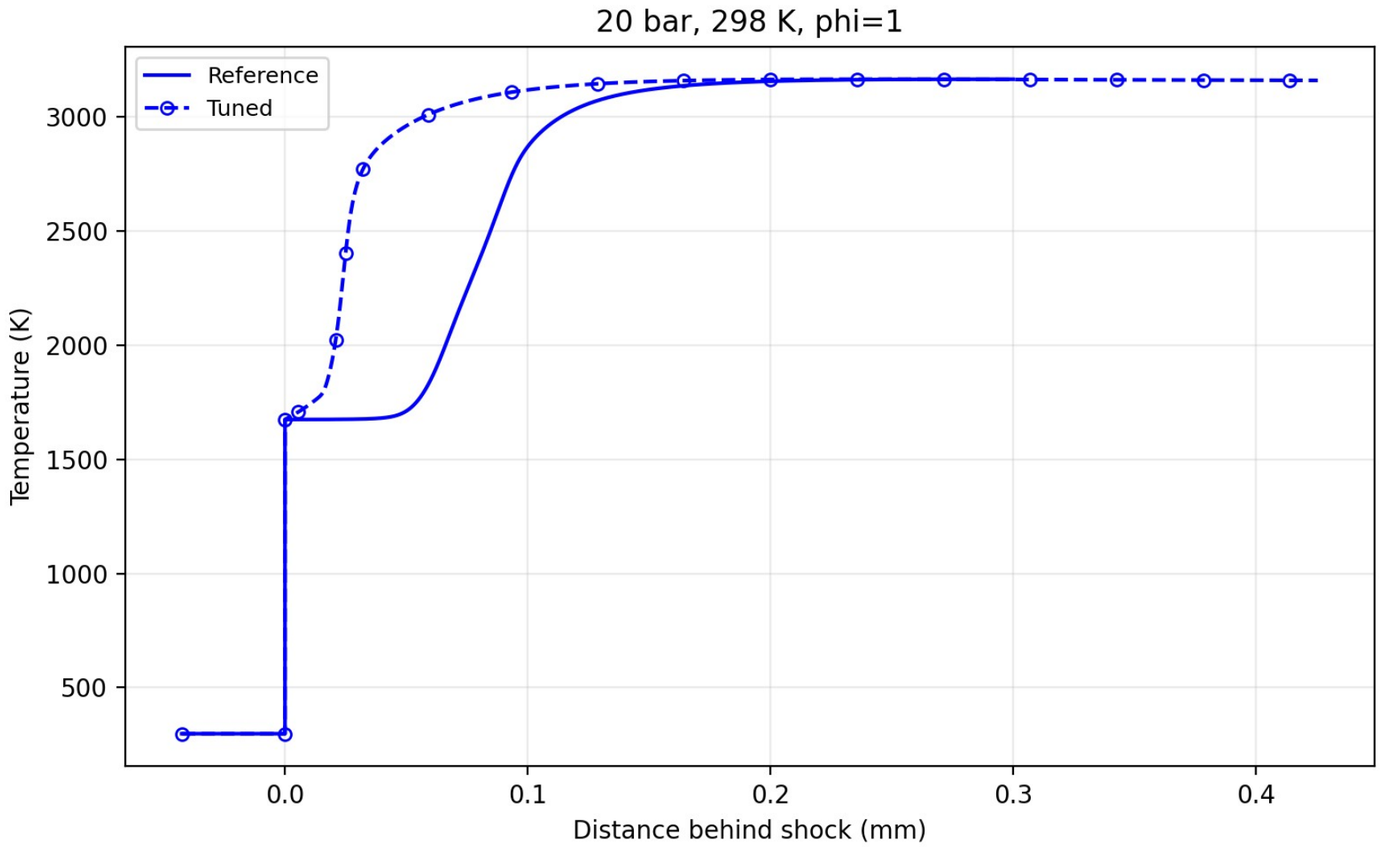
ΔT = +1.733 K

|ln(ℓind,model / ℓind,FFCM2)| = 0.0859669

|ln(ℓexo,model / ℓexo,FFCM2)| = 0.764486

Case 10: 20 bar, φ = 1.0

Ethylene–air, initial temperature 298 K



Reported numerical comparison

Quantity	FFCM2	Tuned	Error
T (K)	3164.69	3159.58	-0.16%
ℓind (m)	8.8824e-5	2.4049e-5	-72.93%
ℓexo (m)	4.0357e-5	7.4619e-6	-81.51%

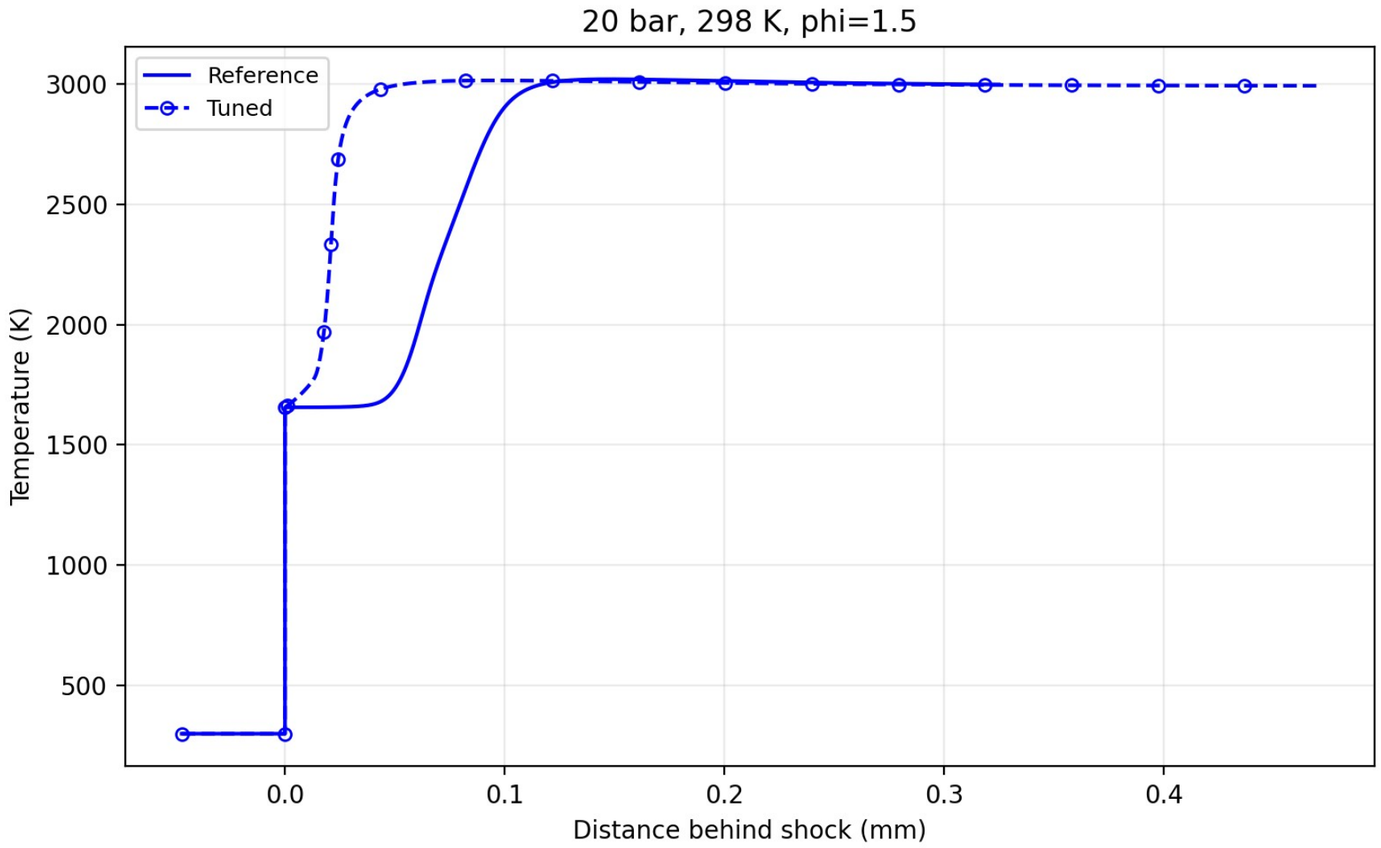
ΔT = -5.111 K

|ln(ℓind,model / ℓind,FFCM2)| = 1.30657

|ln(ℓexo,model / ℓexo,FFCM2)| = 1.68795

Case 11: 20 bar, φ = 1.5

Ethylene–air, initial temperature 298 K



Reported numerical comparison

Quantity	FFCM2	Tuned	Error
T (K)	2998.07	2992.75	-0.18%
ℓind (m)	6.4302e-5	2.1132e-5	-67.14%
ℓexo (m)	4.7288e-5	5.7344e-6	-87.87%

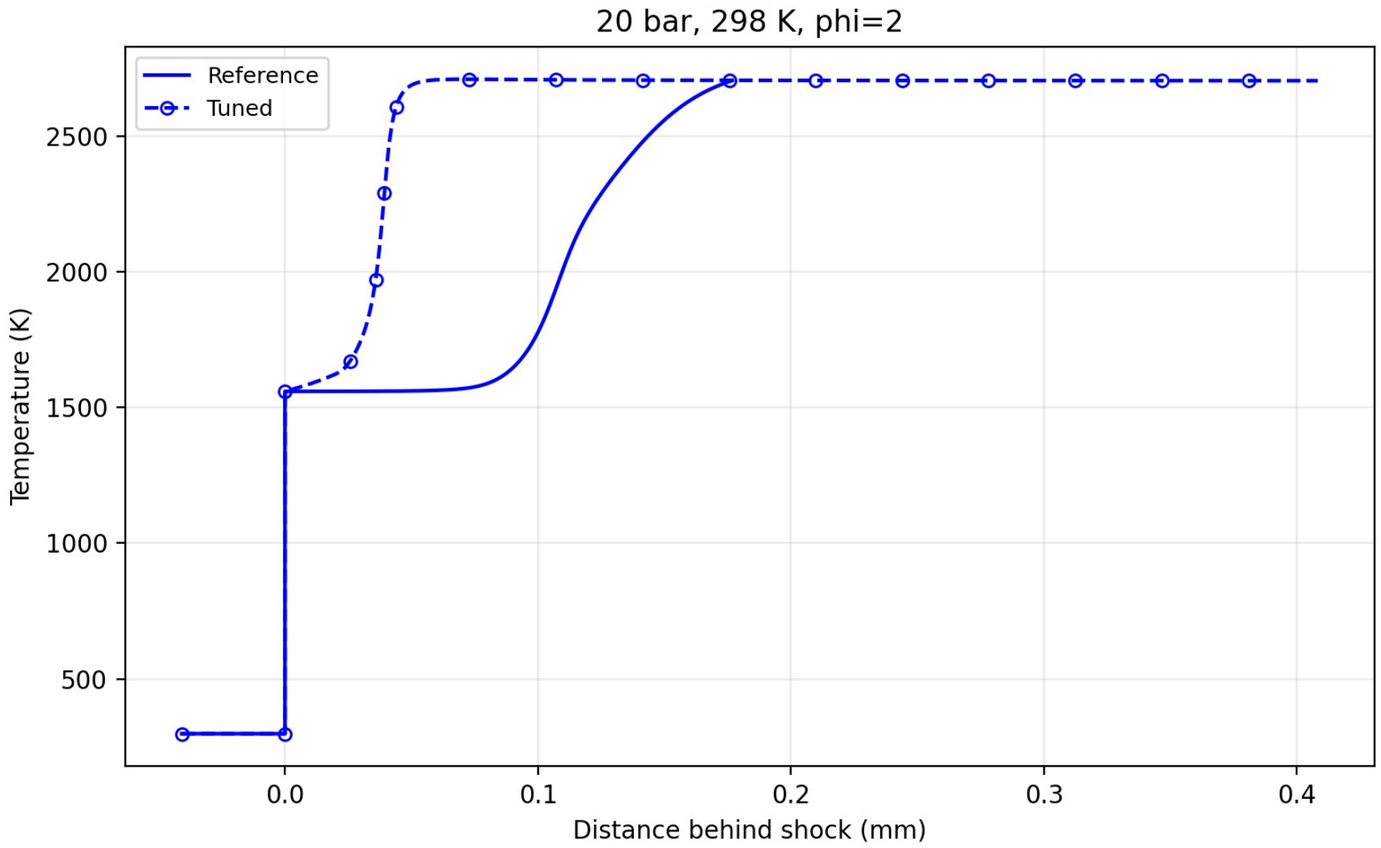
ΔT = -5.320 K

|ln(ℓind,model / ℓind,FFCM2)| = 1.11279

|ln(ℓexo,model / ℓexo,FFCM2)| = 2.10979

Case 12: 20 bar, φ = 2.0

Ethylene–air, initial temperature 298 K



Reported numerical comparison

Quantity	FFCM2	Tuned	Error
T (K)	2705.06	2704.03	-0.04%
ℓind (m)	1.1070e-4	3.9561e-5	-64.26%
ℓexo (m)	4.2647e-5	6.7873e-6	-84.09%

ΔT = -1.029 K

|ln(ℓind,model / ℓind,FFCM2)| = 1.029

|ln(ℓexo,model / ℓexo,FFCM2)| = 1.83791